



Chapter 3: Econometric Modelling

Set up the R programming environment for Chapter 3.

► Show R Code

```
Year Number of Car Registration ('000) BLR (%) Unemployment
Rate (%)
1 1969                21.60      9.1
7.6
2 1970                23.10      9.2
7.5
3 1971                25.80      9.2
7.5
4 1972                29.40      8.9
7.4
5 1973                31.20      9.9
7.2
6 1974                41.61     11.2
7.2
Total Population  CPI Total Export Income      GDP
1      10061684 -0.41      1.73    264 17817535563
2      10306508  1.84      1.76    264 18884189212
3      10552557  1.61      1.72    264 20779153503
4      10801619  3.23      1.82    264 22729992899
5      11062664 10.56      3.18    264 25389647919
6      11335187 17.33      4.59    362 27501726906
```

► Show R Code

```
'data.frame':  54 obs. of  9 variables:
 $ Year                : int  1969 1970 1971 1972
1973 1974 1975 1976 1977 1978 ...
 $ Number of Car Registration ('000): num  21.6 23.1 25.8 29.4
31.2 ...
 $ BLR (%)              : num  9.1 9.2 9.2 8.9 9.9
11.2 10.3 10 9.5 9.3 ...
 $ Unemployment Rate (%) : num  7.6 7.5 7.5 7.4 7.2
7.2 7.4 6.1 6.1 5.4 ...
 $ Total Population      : int  10061684 10306508
10552557 10801619 11062664 11335187 11617947 11910081 12218924
12543892 ...
 $ CPI                   : num  -0.41 1.84 1.61 3.23
10.56 ...
 $ Total Export          : num  1.73 1.76 1.72 1.82
3.18 4.59 4.24 5.73 6.59 8.02 ...
 $ Income                : int  264 264 264 264 264
362 362 505 505 505 ...
 $ GDP                   : num  1.78e+10 1.89e+10
2.08e+10 2.27e+10 2.54e+10 ...
```

1 Introduction

1.1 Definition and Concepts

Econometric modelling is a branch of economics that employs statistical methods to analyse economic phenomena, forecast future trends, and evaluate economic policies. It combines **economic theory**, **mathematics**, and **statistical techniques** to quantify and understand the relationships between economic variables.

Note

Econometric models go beyond univariate time series methods — they explain *why* a variable behaves as it does by relating it to other economic variables (its *determinants*). This gives forecasts a causal interpretation, which is valuable for policy analysis.

1.2 Purpose

Econometric models are used to:

- Test economic theories and hypotheses.
- Predict future economic variables and trends.
- Assess the impact of policy changes or external shocks on the economy.
- Make informed decisions in business, finance, and public policy.

1.3 Applications of Econometric Modelling

Application Area	Example
Macroeconomic Forecasting	Central banks forecast GDP growth, inflation, unemployment rates, and interest rates using large structural models.
Financial Markets	The Capital Asset Pricing Model (CAPM) and Arbitrage Pricing Theory (APT) estimate expected asset returns based on risk.
Labour Economics	Researchers use difference-in-differences and regression discontinuity designs to evaluate minimum wage laws.
Health Economics	Instrumental variables and propensity score matching address endogeneity in evaluating healthcare interventions.
Environmental Economics	Panel data and spatial econometrics assess the effectiveness of

	environmental regulations on economic growth.
Marketing & Consumer Behaviour	Discrete choice models and time series analysis forecast consumer demand and advertising effectiveness.

2 Basic Structure of Econometric Model

2.1 General Form

The variables used in econometric models are:

- **Dependent variable** (y_t): the variable we want to explain or forecast.
- **Independent (explanatory) variables** ($x_{1t}, x_{2t}, \dots, x_{mt}$): the factors that help explain y_t .

The general functional form is:

$$y_t = f(x_{1t}, x_{2t}, \dots, x_{mt}) \quad (1)$$

Equation (1) states that the level of y_t is influenced by the behaviour of variables $x_{1t}, x_{2t}, \dots, x_{mt}$, and the relationship is established from historical data.

2.2 Linear Multiple Regression Model (No Lags)

The standard multiple linear regression model is:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \dots + \beta_m x_{mt} + \varepsilon_t \quad (2)$$

In compact summation form:

$$y_t = \beta_0 + \sum_{i=1}^m \beta_i x_{it} + \varepsilon_t \quad (3)$$

where:

- β_0 is the intercept (constant term),
- β_i is the coefficient for the i -th independent variable,
- ε_t is the error (disturbance) term at time t ,
- The x_{it} matrix is assumed non-stochastic (fixed), and y_t is a random variable.

2.3 General Model with Lag Variables

A more general model, the **Autoregressive Distributed Lag (ADL)** model, allows both lagged values of the dependent variable and lagged values of the independent variables:

$$y_t = \beta_0 + \beta_1 x_{1t} + \dots + \beta_m x_{mt} + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \omega_{11} x_{1(t-1)} + \dots + \omega_{pm} x_{m(t-j)} + \varepsilon_t \quad (4)$$

Note

Including lagged values of y_t in the model is one of the most effective remedies for serial correlation. The ADL model is the foundation of the **General-to-Specific** modelling strategy covered later in this chapter.

3 Fundamental of OLS Technique

3.1 Ordinary Least Squares (OLS)

Ordinary Least Squares (OLS) is the standard method for estimating the parameters of a linear regression model. Its main objective is to **minimise the sum of squared errors**:

$$\text{Minimise } \sum_{t=1}^n \varepsilon_t^2 = \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

where $\varepsilon_t = y_t - \hat{y}_t$ is the residual at time t .

3.2 Deriving the OLS Estimators (Simple Regression)

Consider a regression model with one independent variable:

$$y_t = \beta_0 + \beta_1 x_{1t} + \varepsilon_t \quad (5)$$

The fitted equation is:

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1t} \quad (6)$$

The residual is:

$$\varepsilon_t = y_t - \hat{y}_t = y_t - (\hat{\beta}_0 + \hat{\beta}_1 x_{1t}) \quad (7, 8)$$

The sum of squared errors to minimise:

$$\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n \left[y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t} \right]^2 \quad (9)$$

Partial differentiation with respect to β_0 , equating to zero:

$$\frac{\partial \sum_{t=1}^n \varepsilon_t^2}{\partial \beta_0} = 2 \sum_{t=1}^n (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t}) (-1) = 0 \quad (10)$$

This simplifies (through equations 11–12) to the **first normal equation**:

$$\sum_{t=1}^n y_t = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{t=1}^n x_{1t} \quad (13)$$

Partial differentiation with respect to β_1 , equating to zero:

$$\frac{\partial \sum_{t=1}^n \varepsilon_t^2}{\partial \beta_1} = \sum_{t=1}^n (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t}) (-x_{1t}) = 0 \quad (14)$$

This gives the **second normal equation**:

$$\sum_{t=1}^n x_{1t}y_t = \hat{\beta}_0 \sum_{t=1}^n x_{1t} + \hat{\beta}_1 \sum_{t=1}^n x_{1t}^2 \quad (16)$$

Solving these two normal equations simultaneously yields the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$.

Note

In practice, R's `lm()` function performs all of these calculations internally using matrix algebra. Understanding the derivation helps you interpret what the function is doing under the hood and why the assumptions matter.

4 Issues in the Econometric Construction

4.1 Variable Selection Criteria

Variables included in the model must satisfy two conditions:

1. **Statistical significance** — the variable must have a meaningful statistical relationship with y_t .
2. **Theoretical (logical) justification** — the relationship must be supported by economic theory.

Variable selection is the procedure of choosing a subset (from all possible subsets) of independent variables. The common criterion is to **minimise the squared error** while maintaining theoretical coherence.

4.1.1 Steps for Model Specification

Step	Description
Step 1	Identify the appropriate economic theory to select independent variables that explain the dependent variable.
Step 2	Ensure sufficient data are available. Rule of thumb: at least 5 observations per independent variable .
Step 3	Examine the relationships among variables using historical data to assess the general fitness of the model.

4.2 Model Cost

- As model complexity increases, data requirements and computational costs increase.
- More sophisticated models require more time, specialised software, and

- greater expertise.
- A parsimonious model is preferred when it achieves comparable predictive performance at lower cost.

4.3 Model Complexity

- Simple linear models are easy to interpret and visualise.
- Models with many independent variables are difficult to envisage and may overfit the data.
- **Principle of parsimony:** if two models have the same explanatory power, prefer the simpler one.

4.4 Future Values of Independent Variables

A key practical challenge in multi-variable forecasting:

$$\hat{y}_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{1(t+1)} + \hat{\beta}_2 x_{2(t+1)}$$

To forecast y_{t+1} , the analyst must also have (or forecast) $x_{1(t+1)}$ and $x_{2(t+1)}$.

Note

Challenge: If the future values of independent variables are themselves estimated with error, those errors propagate into the forecast of y . This is a key limitation of econometric forecasting compared to univariate methods — more variables means more sources of forecast error.

5 Assumptions

5.1 Assumptions Pertaining to the Model

For OLS estimates to have desirable properties (unbiasedness, efficiency), the following model-level assumptions must hold:

1. **Linearity** — y_t is a linear function of the independent variables.
2. **Correct specification** — all relevant variables are included; no irrelevant variables are present.
3. **No multicollinearity** — the x_t 's must be mutually independent.
4. **Sufficient sample size** — the number of observations n must exceed the number of regressors m .

5.2 Assumptions Pertaining to the Error Term

1. **IID errors** — the error terms ε_t are identically and independently distributed.
2. **Homoscedasticity** — $\text{Var}(\varepsilon_t) = \sigma^2$ is constant for all t .
3. **No autocorrelation** — $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for all $t \neq s$.
4. **Exogeneity** — the independent variables are uncorrelated with the error

terms.

5. **Normality** — $\varepsilon_t \sim N(0, \sigma^2)$.

5.3 Specification Error

A model is **correctly specified** if it includes the correct set of independent variables. Common misspecification situations are:

Type	Description
Under-specification	Relevant variables are omitted from the model.
Over-specification	Irrelevant variables are included in the model.
Wrong functional form	Linear model used when the true relationship is non-linear.
Poor residuals	Diagnostic checks on residuals indicate systematic patterns.

6 Model Estimation Procedure

6.1 Two Main Strategies

Two main strategies exist for estimating an econometric model:

Strategy	Description
General-to-Specific	Start with the most general model and systematically remove insignificant variables. Also called <i>backward stepwise</i> .
Specific-to-General	Start with a simple model and add variables one at a time. Also called <i>forward stepwise</i> .

6.2 General-to-Specific (Backward Stepwise)

The General-to-Specific approach is based on the **principle of parsimony**:

If there are two or more competing explanations of the same phenomenon, each having the same explanatory power, choose the simpler one.

Procedure:

1. Formulate a sufficiently large **unrestricted (general) model** — the ADL model — including all candidate independent variables and their lags:

$$y_t = \beta_0 + \sum_{m=1}^k \beta_m x_{mt} + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{m=1}^k \sum_{j=1}^q \omega_{mj} x_{m(t-j)} + \varepsilon_t$$

2. Apply diagnostic tests at each stage.
3. Drop the **most insignificant** variable (highest p -value) at each step — one at a time.
4. A variable may also be dropped if it registers the **wrong sign** contrary to theoretical expectations.
5. Repeat until all remaining variables are significant and pass all diagnostics.

Note

Why drop one variable at a time? The significance of one variable depends on what other variables are in the model. Dropping multiple variables simultaneously can lead to incorrect conclusions about which variables truly matter.

Practical considerations for the general model:

- Maintain a small number of explanatory variables to control cost, avoid multicollinearity, and keep the model interpretable.
- If multicollinearity exists, the offending variable (usually with the highest VIF) is a candidate for removal.

6.3 Specific-to-General (Forward Stepwise)

Procedure:

1. Start with the simplest model, e.g.:

$$y_t = \beta_0 + \beta_1 x_{1t} + \varepsilon_t \quad \text{or} \quad y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$$

2. If the model is inadequate or mis-specified, add the next most promising variable (based on correlation with y_t).
3. Repeat until a well-specified model is obtained.

Note

In this course, we primarily use the **General-to-Specific** approach, as illustrated in the worked example with Malaysian car registration data.

7 Statistical Validation and Testing Procedure

Failure to satisfy any of the following tests indicates that a model assumption is violated. Remedial action should then be taken: reformulate the model, include new variables, obtain more data, or apply variable transformations.

Test	Purpose	Decision Rule
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F-test	Overall model fitness	If $p < 0.05$, at least one variable is significant
t-test	Significance of each coefficient	If $p < 0.05$, retain the variable
Adjusted R^2	Goodness of fit	Higher is better; preferred over R^2
Breusch-Pagan	Heteroscedasticity	If $p > 0.05$, variances are constant (no problem)
Durbin-Watson	Serial correlation	DW ≈ 2 and $p > 0.05$ means no problem
VIF	Multicollinearity	VIF > 10 indicates a serious problem

7.1 General Fitness (F-Test)

The F-test evaluates the **overall significance** of the model:

- H_0 : All slope coefficients are zero ($\beta_1 = \beta_2 = \dots = \beta_m = 0$).
- H_1 : At least one coefficient is non-zero.

Decision: If the p -value from the F-test is less than 0.05, the model has overall fit and at least one variable is significant. Proceed to examine individual coefficients using t-tests. Otherwise, the model is considered poorly fit.

7.2 Regression Coefficients (t-Test)

After confirming the model is overall significant via the F-test, examine each individual coefficient $\beta_1, \beta_2, \dots, \beta_m$:

- H_0 : $\beta_i = 0$ (variable i has no effect on y)
- H_1 : $\beta_i \neq 0$

Decision: If $p < 0.05$, retain the variable. Otherwise, consider removing it from the model.

7.3 Goodness of Fit (R^2 and Adjusted R^2)

The **coefficient of determination** R^2 measures the proportion of total variation in y_t explained by the regression:

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y})^2}$$

$R^2 \in [0, 1]$, where 0 indicates no fit and 1 indicates perfect fit.

Problem with R^2 : It increases mechanically whenever a new variable is added, even if the variable adds no explanatory value. Therefore, use **Adjusted R^2** :

$$\bar{R}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - m - 1}$$

where n is the number of observations and m is the number of independent variables. Adjusted R^2 penalises for adding variables and is the preferred measure of goodness of fit.

7.4 Heteroscedasticity

Homoscedasticity means the variance of the error term is constant:

$$\text{Var}(\varepsilon_t) = \sigma^2 \quad \text{for all } t$$

If this assumption is violated, the errors are **heteroscedastic** — their variance changes over time or across observations. Consequences:

- OLS estimates are no longer the minimum variance (BLUE) estimators.
- Standard errors and inference become unreliable.
- Forecasts become less predictable as the forecast horizon extends.

Breusch-Pagan Test: - H_0 : Error variance is constant (homoscedastic). - H_1 : Error variance is non-constant (heteroscedastic). - **Decision:** If $p > 0.05$, no heteroscedasticity problem. If $p < 0.05$, heteroscedasticity is present — consider transforming the dependent variable (e.g., log transformation).

7.5 Serial Correlation (Autocorrelation)

The error terms must be uncorrelated across time:

$$\text{Cov}(\varepsilon_t, \varepsilon_{t-s}) = 0 \quad \text{for all } s \neq 0$$

The most common form is **first-order autocorrelation**:

$$\varepsilon_t = \rho\varepsilon_{t-1} + \nu_t \quad (17)$$

7.5.1 Consequences of Autocorrelation

By repeated substitution of equation (17):

$$\varepsilon_t = \rho\varepsilon_{t-1} + \nu_t = \rho^k\varepsilon_{t-k} + \sum_{j=0}^{k-1} \rho^j\nu_{t-j} \quad (18)$$

Under perfect correlation ($\rho = 1$):

$$\varepsilon_t = \varepsilon_{t-k} + \sum_{j=1}^{k-1} \nu_{t-j} \quad (19)$$

$$\text{Var}(\varepsilon_t) = \sigma_\varepsilon^2 + k\sigma_\nu^2$$

Since the variance of ε_t **grows with** k , forecast accuracy deteriorates as the forecast horizon increases. When $\rho = 0$, $\text{Var}(\varepsilon_t) = \sigma_\varepsilon^2$ is constant and there is no serial correlation.

7.5.2 Durbin-Watson Test

$$DW = \frac{\sum_{t=2}^n (\varepsilon_t - \varepsilon_{t-1})^2}{\sum_{t=1}^n \varepsilon_t^2} \quad (20)$$

DW Value	Interpretation
$DW < 2$	Positive serial correlation
$DW \approx 2$	No serial correlation
$DW > 2$	Negative serial correlation

- **Decision:** If the p -value from the Durbin-Watson test is less than 0.05, serial correlation is present.
- **Remedy:** Include a lag of the dependent variable (y_{t-1}) or a lag of an independent variable. Usually, lag 1 is sufficient to resolve the problem.

7.6 Multicollinearity

Multicollinearity occurs when two or more independent variables are linearly related to each other.

7.6.1 Example

Consider $y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \varepsilon_t$ where $x_{1t} = 2x_{2t}$:

$$y_t = \beta_0 + 2\beta_1 x_{2t} + \beta_2 x_{2t} + \varepsilon_t = \beta_0 + (2\beta_1 + \beta_2)x_{2t} + \varepsilon_t$$

It is impossible to separate the individual effects of x_{1t} and x_{2t} — their contributions are **confounded**.

7.6.2 Consequences

- Standard errors of the coefficients become inflated.
- Individual variables appear insignificant (high p -values) despite the model having a high R^2 .
- Signs of coefficients may be reversed relative to theoretical expectations.

7.6.3 Detection — Variance Inflation Factor (VIF)

$$VIF_i = \frac{1}{1 - R_i^2}$$

where R_i^2 is the R^2 from regressing variable i on all other independent variables.

VIF	Interpretation
$VIF < 5$	No problem
$5 \leq VIF < 10$	Moderate concern
$VIF \geq 10$	Serious multicollinearity — action required

7.6.4 Remedial Actions

1. **Do nothing** if it does not affect model performance or forecasting accuracy.
2. **Drop the offending variable** — choose the one with the least theoretical support or the highest VIF.
3. **Increase the sample size** — more data reduces the impact of multicollinearity.

8 Forecasting Using Econometric Models in R

8.1 Worked Example: Forecasting New Car Registrations in Malaysia

This section demonstrates the complete step-by-step procedure for building an econometric model for forecasting purposes.

Objective: Develop an econometric model to forecast the demand for cars in Malaysia, measured by the **Number of New Car Registrations ('000)**.

8.1.1 Data Overview

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Table 1: First six rows of the economic dataset

Year	Number of Car Registration (^{'000})	BLR (%)	Unemployment Rate (%)	Total Population	Total		Total		GD
					CPI	Export	Income		
1969	21.60	9.1	7.6	10061684	-0.41	1.73	264	1781753556	
1970	23.10	9.2	7.5	10306508	1.84	1.76	264	1888418921	
1971	25.80	9.2	7.5	10552557	1.61	1.72	264	2077915350	
1972	29.40	8.9	7.4	10801619	3.23	1.82	264	2272999289	
1973	31.20	9.9	7.2	11062664	10.56	3.18	264	2538964791	
1974	41.61	11.2	7.2	11335187	17.33	4.59	362	2750172690	

► Show R Code

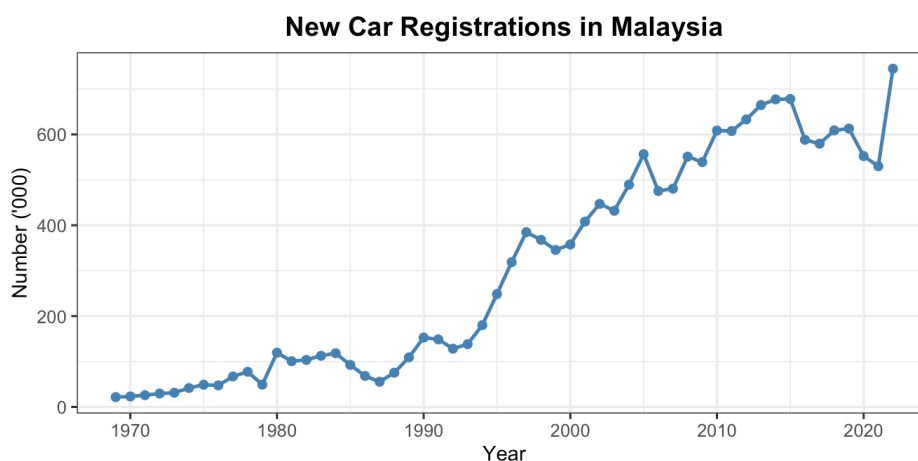


Figure 1: Number of new car registrations in Malaysia, 1969–2022

8.1.2 Variable Names

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```
[1] "Year"                                "Number of Car  
Registration ('000)"  
[3] "BLR (%)"                            "Unemployment Rate (%)"  
[5] "Total Population"                  "CPI"  
[7] "Total Export"                      "Income"  
[9] "GDP"
```

The dataset contains **54 annual observations** from 1969 to 2022. Based on economic theory, the expected relationships between each independent variable and new car registrations are:

Independent Variable	Expected Sign	Rationale
Unemployment Rate (%)	Negative	Higher unemployment reduces household income and car purchases
CPI	Negative	Higher prices reduce real purchasing power
BLR (%)	Negative	Higher lending rates raise financing costs, dampening demand
GDP	Positive	Higher national income supports consumer spending
Total Export	Positive	Export growth reflects economic prosperity
Total Population	Positive	More people, more demand for cars
Income	Positive	Higher per capita income directly boosts car demand

8.1.3 Creating the Lag Variable

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8.2 Model 1 — Full Model (All Variables, No Lag)

The first model includes all seven independent variables. This is the **general model** in the General-to-Specific procedure.

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1(\text{Unemployment}) + \hat{\beta}_2(\text{CPI}) + \hat{\beta}_3(\text{BLR}) + \hat{\beta}_4(\text{GDP}) + \hat{\beta}_5(\text{Export}) + \hat{\beta}_6(\text{Population}) + \hat{\beta}_7(\text{Income})$$

► Show R Code

```
Call:
lm(formula = `Number of Car Registration ('000)` ~
`Unemployment Rate (%)` +
  CPI + `BLR (%)` + GDP + `Total Export` + `Total Population`
+
  Income, data = econ)
```

Residuals:

Min	1Q	Median	3Q	Max
-133.033	-25.599	6.262	18.042	103.360

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.108e+02	1.343e+02	0.825	0.41347
`Unemployment Rate (%)`	-2.070e+01	7.653e+00	-2.705	0.00953
**				
CPI	4.406e-01	2.598e+00	0.170	0.86604
`BLR (%)`	-8.711e+00	6.251e+00	-1.394	0.17016
GDP	-1.601e-09	1.156e-09	-1.385	0.17277
`Total Export`	1.768e+00	3.860e-01	4.579	3.55e-05

`Total Population`	1.454e-05	6.130e-06	2.372	0.02192
*				
Income	3.715e-02	3.401e-02	1.092	0.28039

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 46.27 on 46 degrees of freedom

Multiple R-squared: 0.9673, Adjusted R-squared: 0.9623

F-statistic: 194.5 on 7 and 46 DF, p-value: < 2.2e-16

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--- VIF (Multicollinearity) ---

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`Unemployment Rate (%)`	CPI
3.849078	1.393722
7.000983	
GDP	`Total Export`
414.470181	34.029012
54.943368	
Income	
170.152327	

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--- Durbin-Watson Test (Serial Correlation) ---

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lag	Autocorrelation	D-W Statistic	p-value
1	0.4470231	1.101488	0

Alternative hypothesis: $\rho \neq 0$

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--- Breusch-Pagan Test (Heteroscedasticity) ---

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studentized Breusch-Pagan test

data: model1

BP = 7.8119, df = 7, p-value = 0.3495

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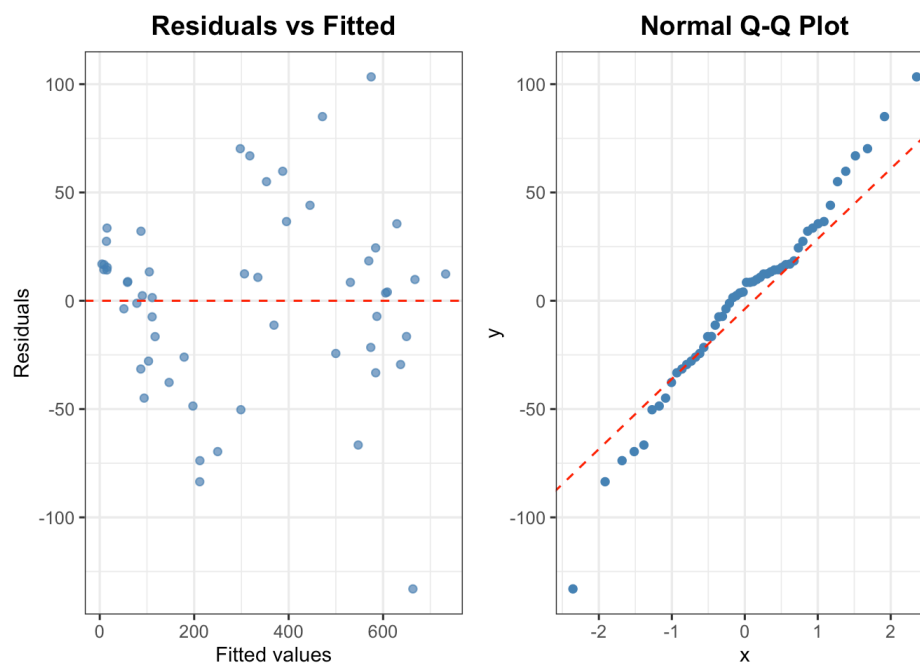


Figure 2: Residual diagnostics for Model 1

Note

Model 1 Interpretation:

- The adjusted R^2 and F-test are highly significant — the variables collectively explain most of the variation in car registrations.
- However, **CPI** and **GDP** registered signs *contrary* to theoretical expectations (positive instead of negative/positive as expected in this context).
- The **VIF of GDP** is extremely high, indicating severe multicollinearity.
- The **Breusch-Pagan** p -value > 0.05 : no heteroscedasticity problem.
- The **Durbin-Watson** test is significant ($p < 0.05$): **serial correlation problem exists**.

Action: Add the lag variable of new car registration to address the serial correlation.

8.3 Model 2 — Add Lag of Dependent Variable

Adding y_{t-1} (lag 1 of car registrations) is the standard remedy for serial correlation.

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1(\text{Unemployment}) + \hat{\beta}_2(\text{CPI}) + \hat{\beta}_3(\text{BLR}) + \hat{\beta}_4(\text{GDP}) + \hat{\beta}_5(\text{Export}) + \hat{\beta}_6(\text{Population}) + \hat{\beta}_7(\text{Income}) + \hat{\phi}_1 y_{t-1}$$

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~  
  `Unemployment Rate (%)` +  
    CPI + `BLR (%)` + GDP + `Total Export` + `Total Population`  
  +  
    Income + lag1, data = econ)
```

Residuals:

Min	1Q	Median	3Q	Max
-83.653	-14.255	0.009	13.669	94.445

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.411e+02	1.094e+02	1.290	0.203828
`Unemployment Rate (%)`	-1.473e+01	6.132e+00	-2.402	0.020587
*				
CPI	3.344e-01	2.154e+00	0.155	0.877331
`BLR (%)`	-4.880e+00	4.990e+00	-0.978	0.333483
GDP	-9.560e-10	9.200e-10	-1.039	0.304454
`Total Export`	1.196e+00	3.232e-01	3.702	0.000593

`Total Population`	3.357e-06	5.464e-06	0.614	0.542141
Income	2.357e-02	2.693e-02	0.875	0.386057
lag1	5.119e-01	9.389e-02	5.453	2.14e-06

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36.48 on 44 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.98, Adjusted R-squared: 0.9764

F-statistic: 269.6 on 8 and 44 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

`Unemployment Rate (%)`		CPI	
`BLR (%)`			
	3.694196		1.493748
7.158650			
	GDP	`Total Export`	`Total
Population`			
	411.683506	37.556426	
67.445528			
	Income	lag1	
	168.195560	18.671873	

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

```
lag Autocorrelation D-W Statistic p-value
1 -0.002775646 1.853213 0.168
Alternative hypothesis: rho != 0
```

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

```
data: model2
BP = 18.234, df = 8, p-value = 0.01954
```

Note

Model 2 Interpretation:

- The DW value improved to approximately 1.85 and $p > 0.05$: **serial correlation problem resolved**.
- Only **Total Export**, **Unemployment Rate**, and the **lag variable** are statistically significant.
- **CPI** and **GDP** still show incorrect signs, and GDP has a very high VIF — multicollinearity persists.

Action: Drop GDP (most insignificant with highest VIF) from the model.

8.4 Model 3 — Remove GDP

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~
`Unemployment Rate (%)` +
  CPI + `BLR (%)` + `Total Export` + `Total Population` +
  Income +
  lag1, data = econ)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-77.268	-16.073	-0.469	11.554	104.394

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.365e+02	1.094e+02	1.248	0.218566
`Unemployment Rate (%)`	-1.304e+01	5.918e+00	-2.204	0.032701

*

CPI	1.965e-02	2.134e+00	0.009	0.992694
`BLR (%)`	-3.492e+00	4.812e+00	-0.726	0.471846
`Total Export`	9.986e-01	2.614e-01	3.820	0.000406

`Total Population`	4.600e-07	4.703e-06	0.098	0.922522
Income	-3.264e-03	7.613e-03	-0.429	0.670121
lag1	5.248e-01	9.316e-02	5.633	1.09e-06

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36.51 on 45 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9795, Adjusted R-squared: 0.9763

F-statistic: 307.4 on 7 and 45 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

	CPI
`Unemployment Rate (%)`	
`BLR (%)`	
3.434997	1.464204
6.645678	
`Total Export`	
Income	
24.520363	49.886310
13.422505	
lag1	
18.349569	

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

lag	Autocorrelation	D-W Statistic	p-value
1	0.03327008	1.751804	0.084

Alternative hypothesis: rho != 0

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

data: model3

BP = 14.423, df = 7, p-value = 0.04416

Note

Model 3 Interpretation:

- The model improved with higher R^2 and no serial correlation problem.
- However, **CPI** and **Income** registered incorrect signs.
- **Income** is the most insignificant variable.

Action: Drop Income from the model.

8.5 Model 4 — Remove Income

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~  
  `Unemployment Rate (%)` +  
    `BLR (%)` + `Total Export` + `Total Population` + CPI +  
  lag1,  
  data = econ)
```

Residuals:

Min	1Q	Median	3Q	Max
-79.452	-18.570	-0.189	12.885	99.914

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.610e+02	9.250e+01	1.740	0.088530
•				
`Unemployment Rate (%)`	-1.420e+01	5.219e+00	-2.721	0.009169
**				
`BLR (%)`	-3.609e+00	4.762e+00	-0.758	0.452389
`Total Export`	9.939e-01	2.588e-01	3.840	0.000374

`Total Population`	-8.518e-07	3.540e-06	-0.241	0.810923
CPI	-1.456e-01	2.080e+00	-0.070	0.944521
lag1	5.297e-01	9.163e-02	5.781	6.17e-07

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36.19 on 46 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9794, Adjusted R-squared: 0.9768

F-statistic: 365.1 on 6 and 46 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

`Unemployment Rate (%)`		`BLR (%)`		`Total
Export`				
	2.720139		6.624291	
24.477286				
`Total Population`			CPI	

```
lag1
28.777339      1.416478
18.072527
```

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

```
lag Autocorrelation D-W Statistic p-value
1      0.04457492      1.745045    0.122
Alternative hypothesis: rho != 0
```

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

```
data: model4
BP = 13.282, df = 6, p-value = 0.03877
```

Note

Model 4 Interpretation:

- The model produces a high R^2 but most variables remain insignificant — a hallmark of multicollinearity.
- **Total Population** has an incorrect sign (negative, contrary to the expected positive relationship) and is insignificant.

Action: Remove Total Population from the model.

8.6 Model 5 — Remove Total Population

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~
`Unemployment Rate (%)` +
`BLR (%)` + `Total Export` + CPI + lag1, data = econ)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-81.417 -17.893   0.529  12.971  98.928
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    146.07589    68.07994   2.146  0.0371 *
`Unemployment Rate (%)` -13.82127    4.92741  -2.805  0.0073
```

```

**
`BLR (%)`          -3.49417    4.69007  -0.745    0.4600
`Total Export`     0.95831    0.21034   4.556  3.71e-05
***
CPI                0.03282    1.92420    0.017    0.9865
lag1               0.52062    0.08273    6.293  9.67e-08
***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 35.82 on 47 degrees of freedom
 (1 observation deleted due to missingness)
 Multiple R-squared: 0.9794, Adjusted R-squared: 0.9772
 F-statistic: 447.1 on 5 and 47 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

	`Unemployment Rate (%)`	`BLR (%)`	`Total Export`
	2.474048	6.558008	
16.497116			
	CPI	lag1	
	1.236596	15.033042	

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

lag	Autocorrelation	D-W Statistic	p-value
1	0.05141295	1.734915	0.11

Alternative hypothesis: rho != 0

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

data: model5
 BP = 12.122, df = 5, p-value = 0.03315

Note

Model 5 Interpretation:

- The model still shows evidence of multicollinearity: high R^2 but insignificant variables.
- **CPI** is the most insignificant variable.

8.7 Model 6 — Remove CPI

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~
  `Unemployment Rate (%)` +
    `BLR (%)` + `Total Export` + lag1, data = econ)
```

Residuals:

Min	1Q	Median	3Q	Max
-81.403	-17.873	0.738	13.383	98.933

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	146.08660	67.36439	2.169	0.03510 *
`Unemployment Rate (%)`	-13.82329	4.87442	-2.836	0.00667
**				
`BLR (%)`	-3.47933	4.56045	-0.763	0.44924
`Total Export`	0.95889	0.20540	4.669	2.47e-05

lag1	0.52038	0.08062	6.455	5.04e-08

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 35.45 on 48 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9794, Adjusted R-squared: 0.9777

F-statistic: 570.8 on 4 and 48 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

`Unemployment Rate (%)`		`BLR (%)`		`Total
Export`				
	2.472613		6.332415	
16.065072				
	lag1			
	14.580335			

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

lag	Autocorrelation	D-W Statistic	p-value
1	0.05149487	1.734734	0.156

Alternative hypothesis: $\rho \neq 0$

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

data: model6

BP = 11.858, df = 4, p-value = 0.01844

Note

Model 6 Interpretation:

- After removing CPI, the model improved significantly: high R^2 with three significant variables.
- The Durbin-Watson value confirms **no serial correlation problem**.
- However, VIF values for **Total Export** and the lag variable indicate multicollinearity still exists between these two variables.

Action: Remove Total Export from the model.

8.8 Model 7 — Final Model (Remove Total Export)

► Show R Code

Call:

```
lm(formula = `Number of Car Registration ('000)` ~  
  `Unemployment Rate (%)` +  
  `BLR (%)` + lag1, data = econ)
```

Residuals:

Min	1Q	Median	3Q	Max
-86.523	-20.412	-0.523	17.512	180.560

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	264.93628	74.43585	3.559	0.000838

`Unemployment Rate (%)`	-16.68328	5.77138	-2.891	0.005715
**				
`BLR (%)`	-13.84240	4.75453	-2.911	0.005402
**				
lag1	0.78071	0.06949	11.235	3.66e-15

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 42.31 on 49 degrees of freedom

(1 observation deleted due to missingness)
Multiple R-squared: 0.9701, Adjusted R-squared: 0.9682
F-statistic: 529.2 on 3 and 49 DF, p-value: < 2.2e-16

► Show R Code

--- VIF (Multicollinearity) ---

► Show R Code

`Unemployment Rate (%)`		`BLR (%)`
lag1		
	2.433558	4.832143
7.604651		

► Show R Code

--- Durbin-Watson Test (Serial Correlation) ---

► Show R Code

lag	Autocorrelation	D-W Statistic	p-value
1	-0.06392779	1.755684	0.2

Alternative hypothesis: $\rho \neq 0$

► Show R Code

--- Breusch-Pagan Test (Heteroscedasticity) ---

► Show R Code

studentized Breusch-Pagan test

data: model7
BP = 4.4204, df = 3, p-value = 0.2195

► Show R Code

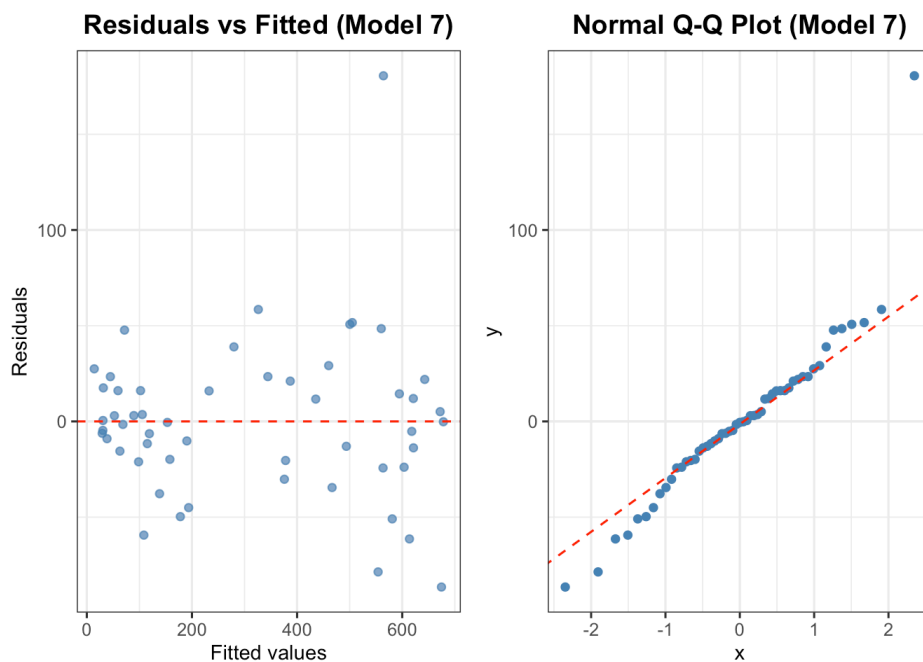


Figure 3: Residual diagnostics for the final model (Model 7)

► Show R Code

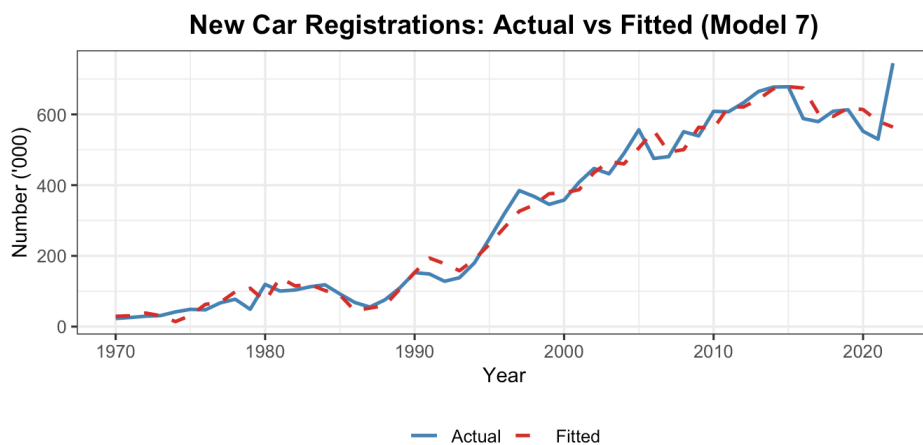


Figure 4: Actual vs fitted values for Model 7 (final model)

Note

Model 7 Interpretation (Final Model):

- The model is overall **highly significant** (F-test $p < 0.05$).
- All three variables — **Unemployment Rate**, **BLR**, and the **lag of car registrations** — are statistically significant.
- **No serial correlation**: $DW \approx 2$, $p > 0.05$.
- **No multicollinearity**: all VIF values are well below 10.
- **No heteroscedasticity**: Breusch-Pagan $p > 0.05$.
- All signs are consistent with economic theory.

8.9 The Final Estimated Equation

The final estimated model is:

$$\hat{y}_t = 264.94 - 16.68 x_{1t} - 13.84 x_{2t} + 0.78 y_{t-1}$$

where:

- $\hat{y}_t$ = estimated number of new car registrations ('000) in year t ,
- x_{1t} = Unemployment Rate (%) in year t ,
- x_{2t} = Base Lending Rate, BLR (%) in year t ,
- y_{t-1} = lag 1 of new car registrations ('000).

8.9.1 Interpretation of Coefficients

Coefficient	Value	Interpretation
Intercept	264.94	Baseline car registrations when all predictors are zero
Unemployment Rate	-16.68	A 1 percentage-point increase in unemployment reduces car demand by approximately 16,680 units
BLR	-13.84	A 1 percentage-point increase in BLR reduces car demand by approximately 13,840 units
Lag of car registrations	+0.78	If car registrations increased by 1,000 last year, demand this year increases by approximately 780 units

All signs are consistent with economic theory: higher unemployment and higher borrowing costs both suppress consumer demand for cars, while past registration levels exhibit strong positive persistence.

8.10 Forecasting with the Final Model

8.10.1 Point Forecast for 2023

Given:

- Unemployment Rate for 2023: $x_{1,2023} = 3.3\%$
- BLR for 2023: $x_{2,2023} = 6.89\%$
- Number of car registrations in 2022: $y_{2022} = 744.78$ ('000)

► Show R Code

Model coefficients:

► Show R Code

```

(Intercept) `Unemployment Rate (%)`
`BLR (%)`
264.9363 -16.6833
-13.8424
lag1
0.7807

```

► Show R Code

Forecast for 2023 (number of new car registrations, '000):
695.96

► Show R Code

Forecast in actual units: 695961

Note

Forecasting caveat: The quality of this forecast depends on how accurately we know (or can forecast) the unemployment rate and BLR for 2023. If those values are themselves uncertain, the forecast error will be larger than the model's in-sample statistics suggest.

8.10.2 Model Selection Summary Table

► Show R Code

Table 2: Summary of all models in the General-to-Specific procedure

Model	Change from Previous	Adj. R ²	DW p-value	BP p-value
Model 1	None (full model)	0.9623	0.000	0.349
Model 2	— (add lag1)	0.9764	0.144	0.020
Model 3	GDP removed	0.9763	0.066	0.044
Model 4	Income removed	0.9768	0.110	0.039
Model 5	Total Population removed	0.9772	0.114	0.033
Model 6	CPI removed	0.9777	0.154	0.018
Model 7	Total Export removed	0.9682	0.200	0.220

9 Summary

This chapter covered the complete framework for **econometric modelling** in time series forecasting. The key takeaways are:

1. **Model Structure:** Econometric models relate a dependent variable to multiple independent (explanatory) variables, grounded in economic theory. The ADL model incorporates both current and lagged variables.
2. **OLS Estimation:** Ordinary Least Squares minimises the sum of squared residuals. Understanding its derivation helps interpret model output and diagnose problems.
3. **Issues in Model Construction:** Variable selection must balance statistical

significance with theoretical justification. Practical challenges include model cost, complexity, and the need for future values of predictors.

4. **Assumptions:** Five key assumptions govern OLS — linearity, correct specification, no multicollinearity, homoscedasticity, and no serial correlation. Violations require diagnostic tests and remedial action.
 5. **Model Estimation Strategies:**
 - **General-to-Specific** (backward): Start with the full model and systematically drop the most insignificant variable at each stage until a parsimonious model is reached.
 - **Specific-to-General** (forward): Start simple and add variables one at a time.
 6. **Statistical Validation:** A valid model must pass six diagnostic tests: F-test, t-tests, adjusted R^2 , Breusch-Pagan (heteroscedasticity), Durbin-Watson (serial correlation), and VIF (multicollinearity).
 7. **Worked Example:** The final model for Malaysian new car registrations identified **Unemployment Rate**, **BLR**, and the **lag of car registrations** as the significant determinants. The model satisfies all diagnostic conditions and produces an interpretable, parsimonious equation.
-

10 References

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